Zecheng Yi

CONTACT INFORMATION

Boston University
Math Department
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RESEARCH INTERESTS

Arithmetic geometry, p-adic Hodge theory, p-adic representation theory.

EDUCATION

Boston University, Boston, MA

Ph.D. Mathematics, September, 2022 –

- Advisor: Jared Weinstein
- Thesis: "Geometric properties of Banach–Colmez spaces"

University of Oxford, The Queen's College, Oxford, UK

M.Sc. Mathematical Science (Part C), September, 2020 – June, 2021

- Dissertation: "Comparing knots and primes using étale fundamental groups" advised by Martin Gallauer.

Johns Hopkins University, Baltimore, MD

B.A./M.A. Mathematics, August, 2016 – May, 2020

- Senior Thesis: "Ring class fields and singular moduli" advised by David Savitt.
- Departmental Honor
- GPA: 3.94/4.0 (Major GPA: 3.98/4.0)

AWARDS

J.J. Sylvester Award for excellence in math, Johns Hopkins University, 2020

PAPERS

- *Banach extensions of almost $\mathbb{C}_p$-representations*, in preparation (available soon).
- *Grassmannians of absolute Banach–Colmez spaces*, arXiv:2610.42110, 2026.
- *On the pro-étale cohomology of quotient stacks of Drinfeld spaces*, arXiv:2510.02699, 2025.

CONFERENCES AND WORKSHOPS

Geometrization of the local Langlands correspondence, Luminy	February, 2026
Complex and p-Adic Simpson Correspondence, Brin Math Research Center	November, 2025
Categorical and Geometric Structures in the Langlands Program, Oaxaca	September, 2025
Period maps: classical and p-adic, Cargèse	September, 2025
Geometry and Representation Theory around the Langlands program, Clermont-Ferrand	July, 2025
Geometric Approaches to the Local Langlands Program, Brin Math Research Center	March, 2025
Multigraded and differential graded methods in commutative algebra, MSRI	August, 2024
Motivic Homotopy Theory, PCMI	July, 2024
Spring School on Non-Archimedean Geometry and Eigenvarieties, Heidelberg	Spring, 2023
JAMI Workshop: Riemann-Roch in Characteristic One, Baltimore	October, 2019

Talks

Continuous cohomology of Steinberg representations, Maine-Québec NT Conference, October 2025

Continuous cohomology of Steinberg representations, Temple GTA Conference, May 2025

Points on eigenvariety with non-critical slopes, JHU Algebraic Number Theory Seminar, May 2020.

Teaching

Boston University, Boston, MA

Course Instructor

- MA 122 Calculus 2 for Social Sciences, Summer 2023
- MA 124 Calculus 2, Summer 2023
- MA 225 Multivariable Calculus, Summer 2025
- MA 581 Probability, Summer 2024, Summer 2026

Teaching Assistant

- MA 123 Calculus 1, Fall 2023, Fall 2024, Fall 2025
- MA 124 Calculus 2, Spring 2026
- MA 213 Basic Statistics 1, Spring 2024
- EK 103 Computational Linear Algebra, Spring 2025

Activities

- Contributor to the Automorphic Project, https://automorphic.jh.edu/
- Reviewer for ZbMath
- Co-organizer of BU Student Number Theory Seminar (2022-2025)

Computer Skills

- Programming: C++ (advanced), C (advanced), Java (advanced), R (proficient), LaTeX(expert), HTML (advanced)
- Languages: English (fluent), Mongolian (native), Chinese (native), French (intermediate), Latin (beginner)